

Lake Powell and Lake Mead Historical and Current Reservoir Data

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Overview

This is an R Markdown document. The document generates plots for Lake Powell and Lake Mead:

1. Annual Lake Powell Release Compared to 7.5 and 8.23 Targets

Data wrangling strategy

1. Import the reservoir elevation-volume curves.
2. Import Daily, Monthly, and Annual Lake Powell and Lake Mead data from the Reclamation's HydroData Web portal https://www.usbr.gov/uc/water/hydrodata/reservoir__data/site__map.html
3. Filter the reservoir data for the appropriate variables/fields.

This is a beginning R-programming effort! There could be lurking bugs or basic coding errors that I am not even aware of. Please report bugs/feedback to me (contact info below)

Requested Citation

David E. Rosenberg (2026), "Lake Powell and Lake Mead Historical and Current Reservoir Data." Utah State University. Logan, Utah. <https://github.com/dzeke/ColoradoRiverCollaborate/tree/main/HistoricalCurrentReservoirData>.

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Figure 1: Annual Lake Powell Release Compared to 7.5 and 8.23 Targets

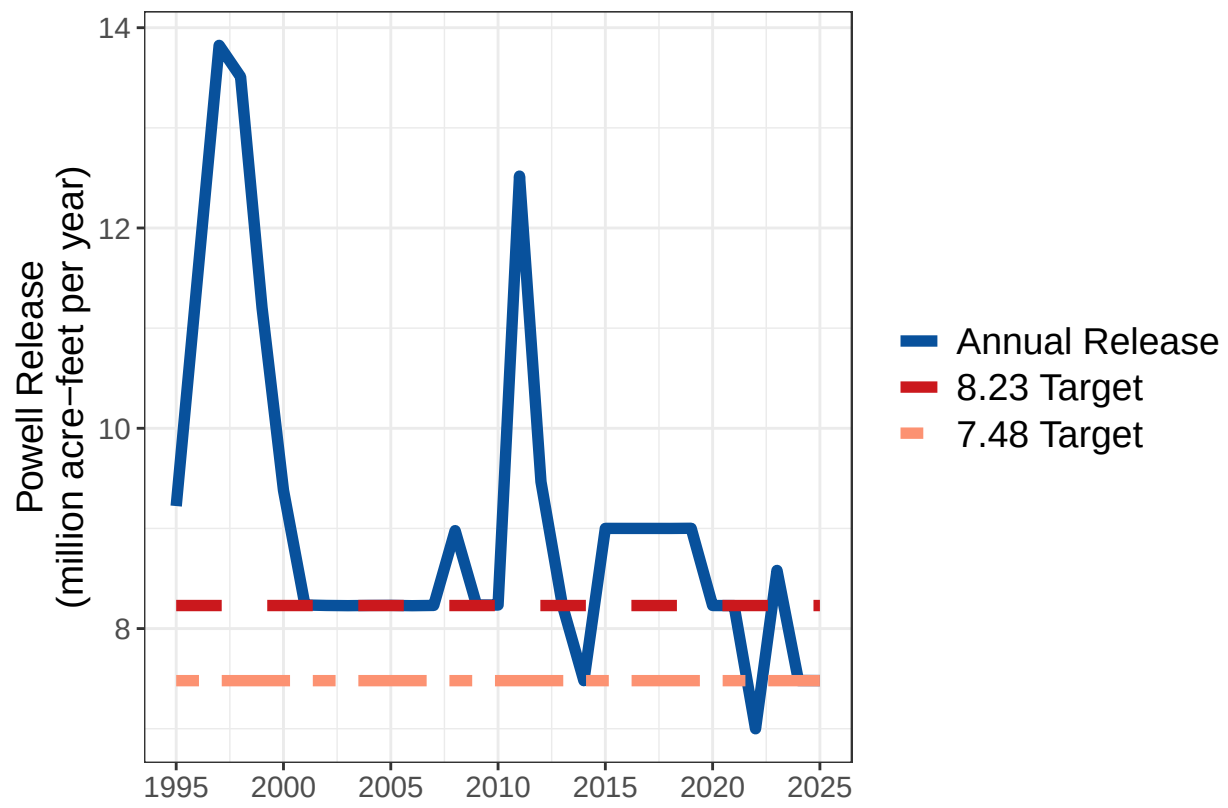


Figure 1: Ten-Year Powell Release Compared to 10-year Targets

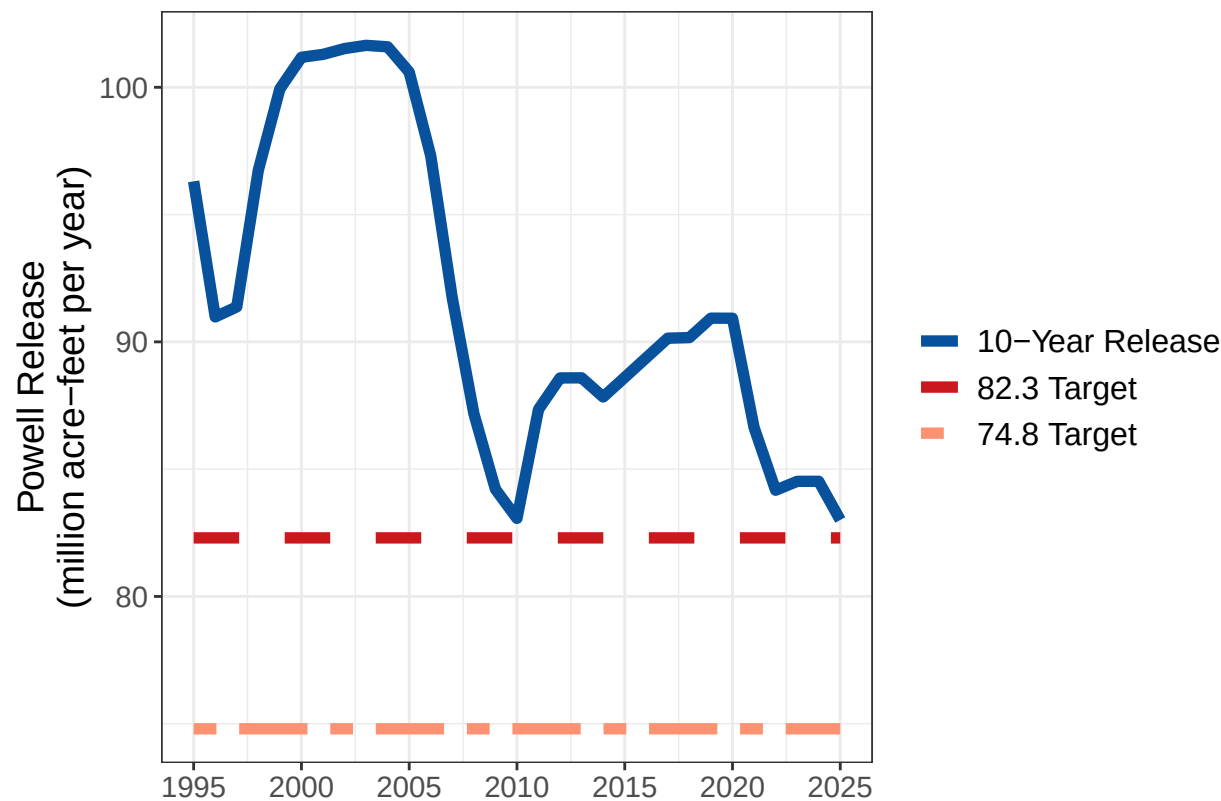


Table 1: Current Storage and Elevations

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ResName	Storage	Pool Elevation
Lake Powell	5.45	3523.29
Lake Mead	7.10	1041.73

Figure 3: Lake Powell, Lake Mead, and Combined Storage

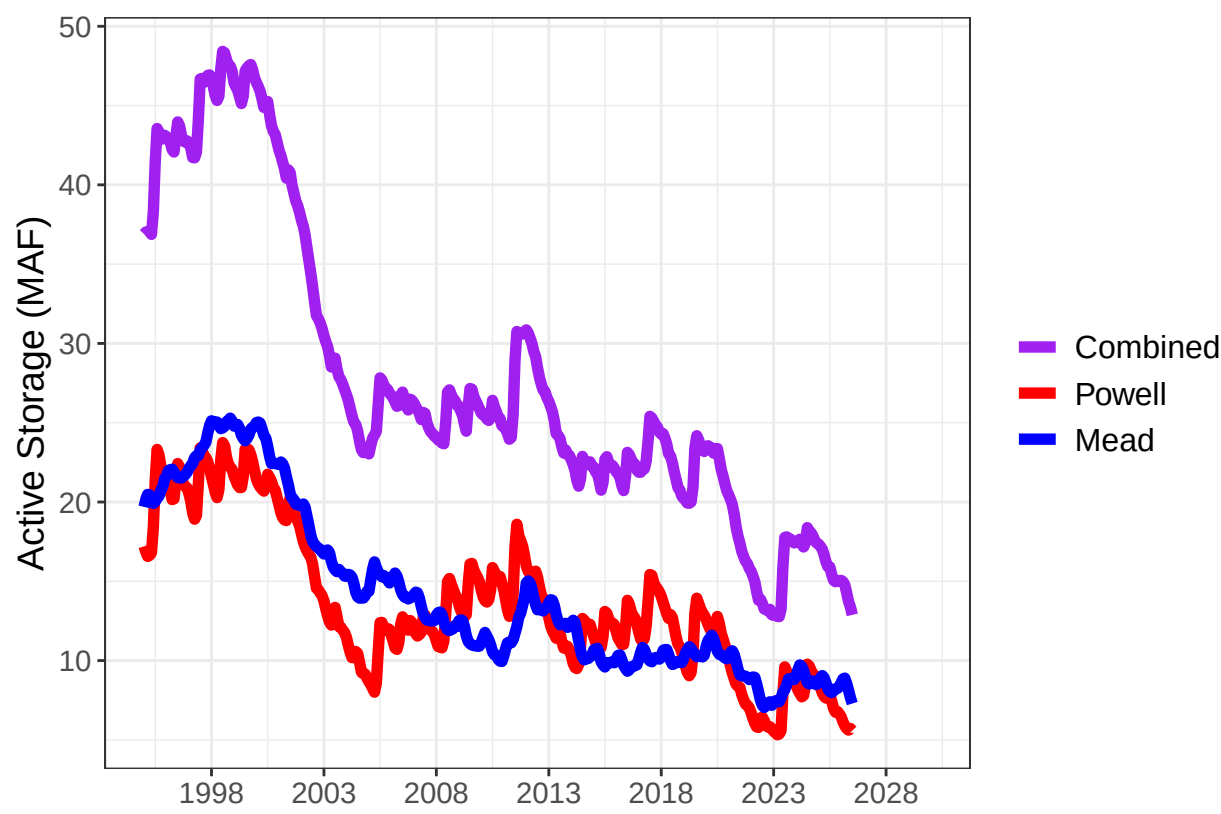


Figure 4: Lake Powell Storage vs Lake Mead Storage

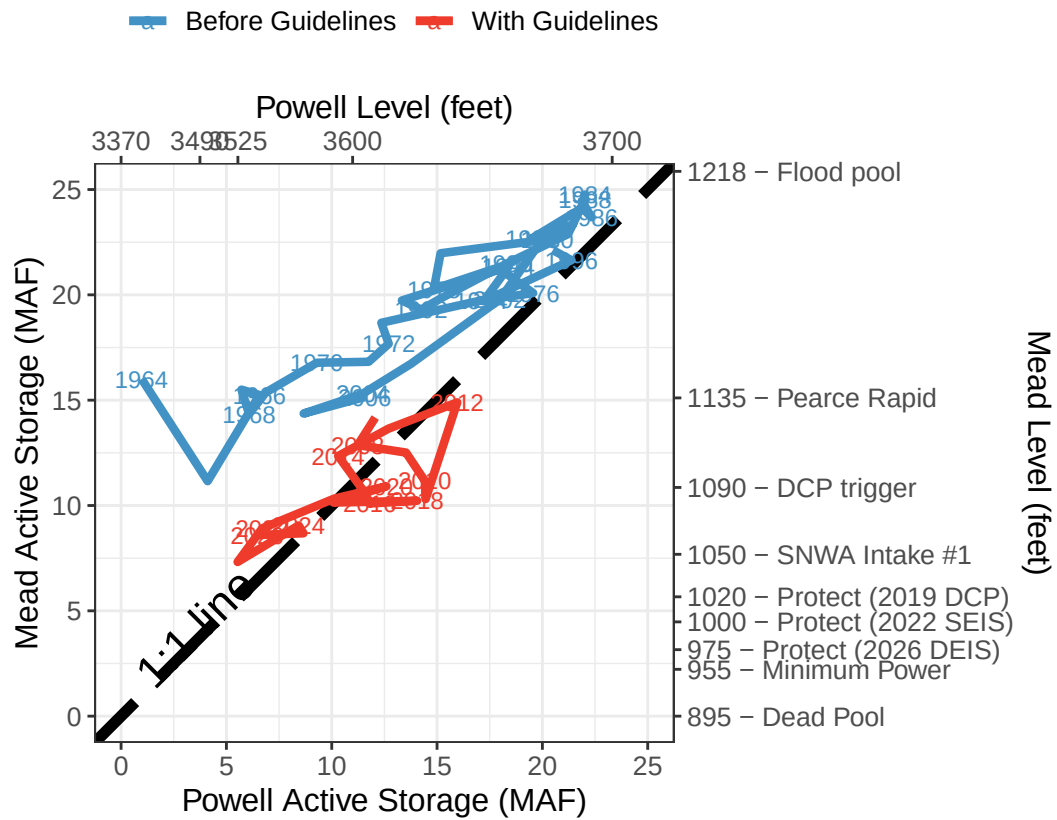


Figure 5: Lake Powell Storage and Elevation over Time.

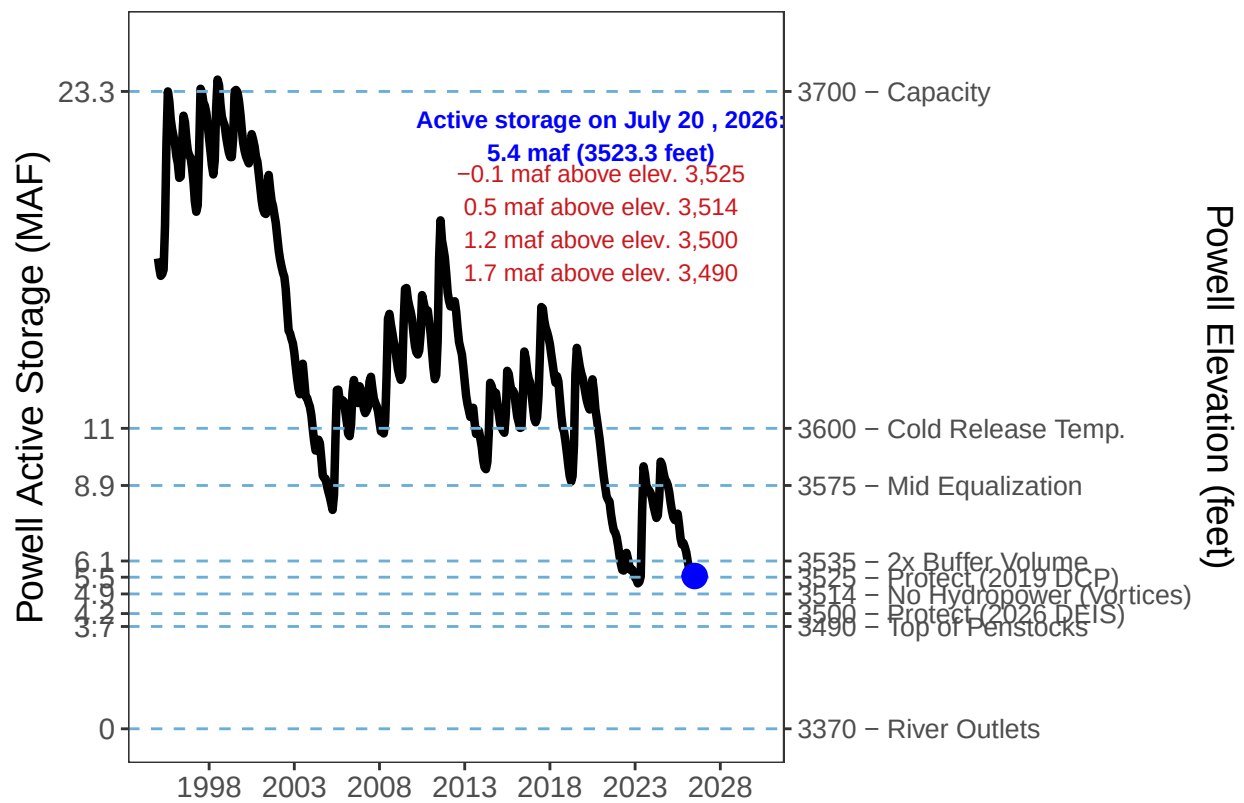


Figure 5B: No Longer Use

Figure 5C: Lake Powell Minimum Drawdown and Catastrophic Elevations - Daily Data

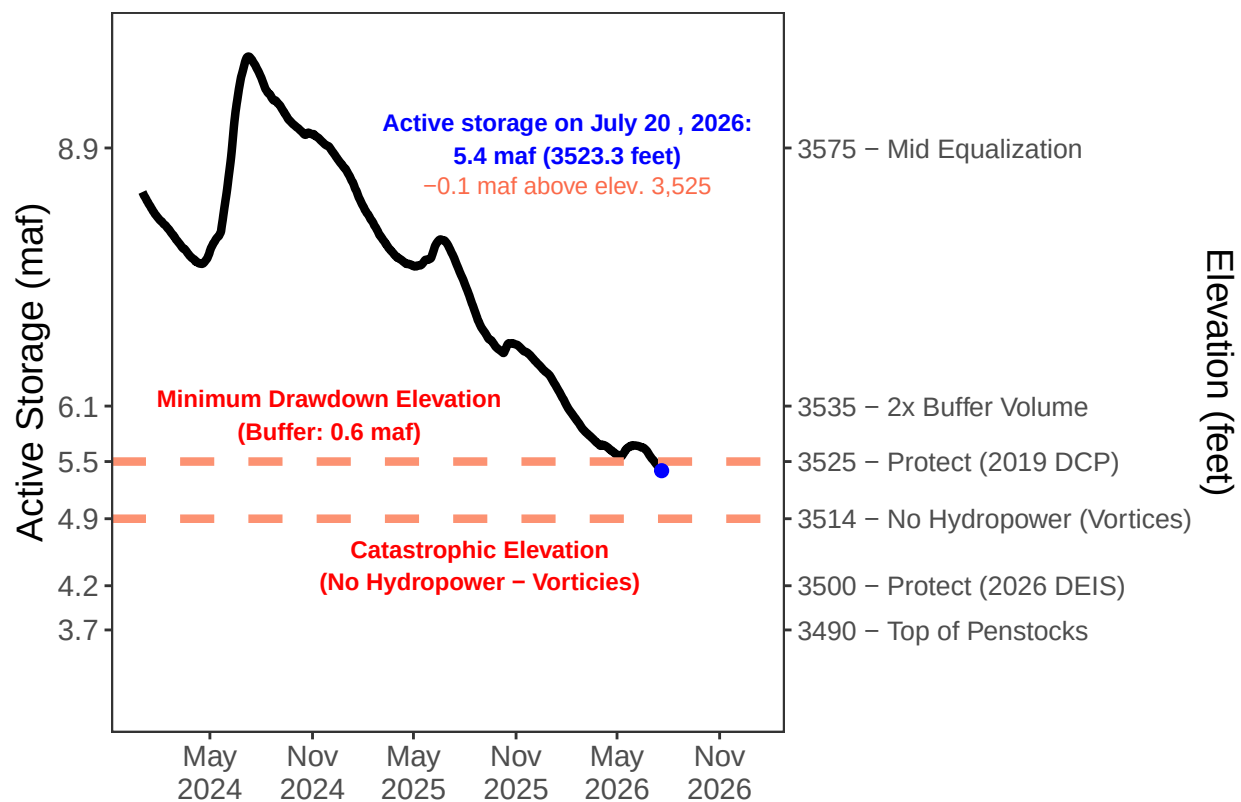


Figure 6: Lake Mead Storage and Elevation over Time.

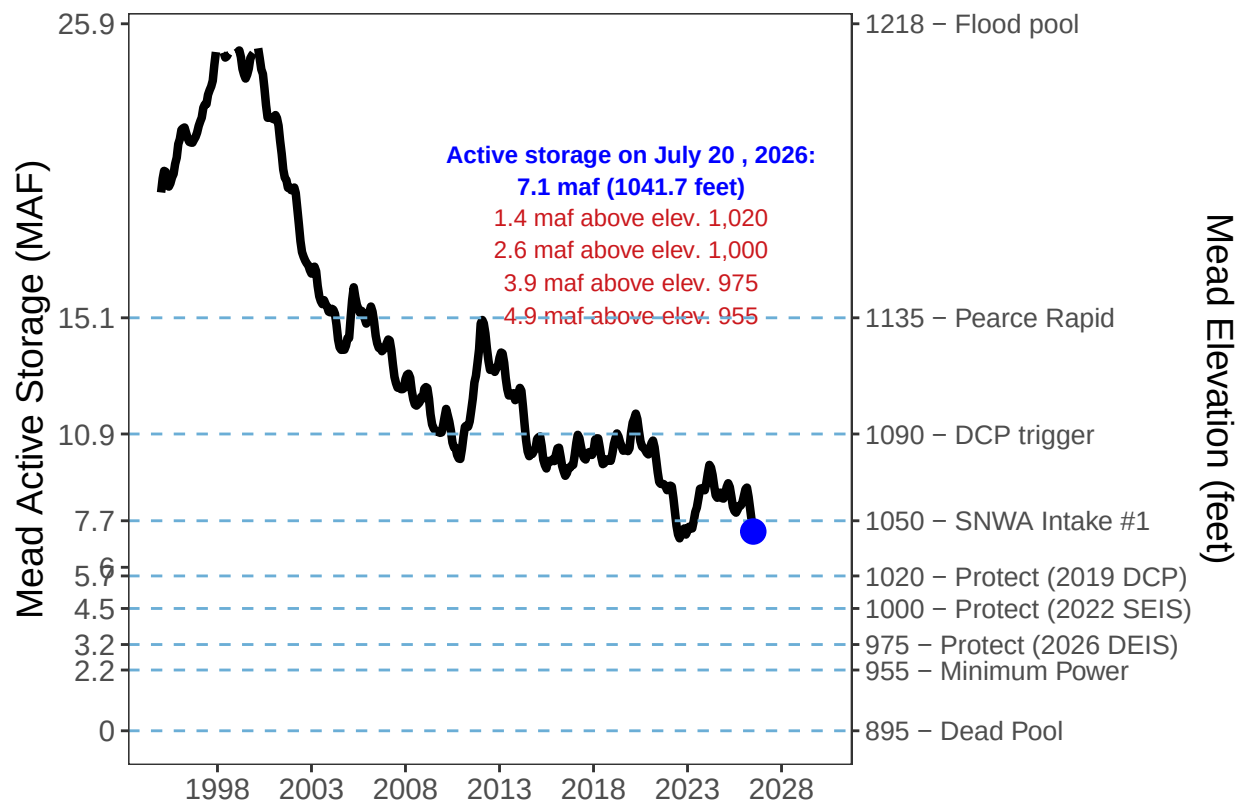


Figure 6B: Lake Powell Daily Inflow and Release Water Year to Date

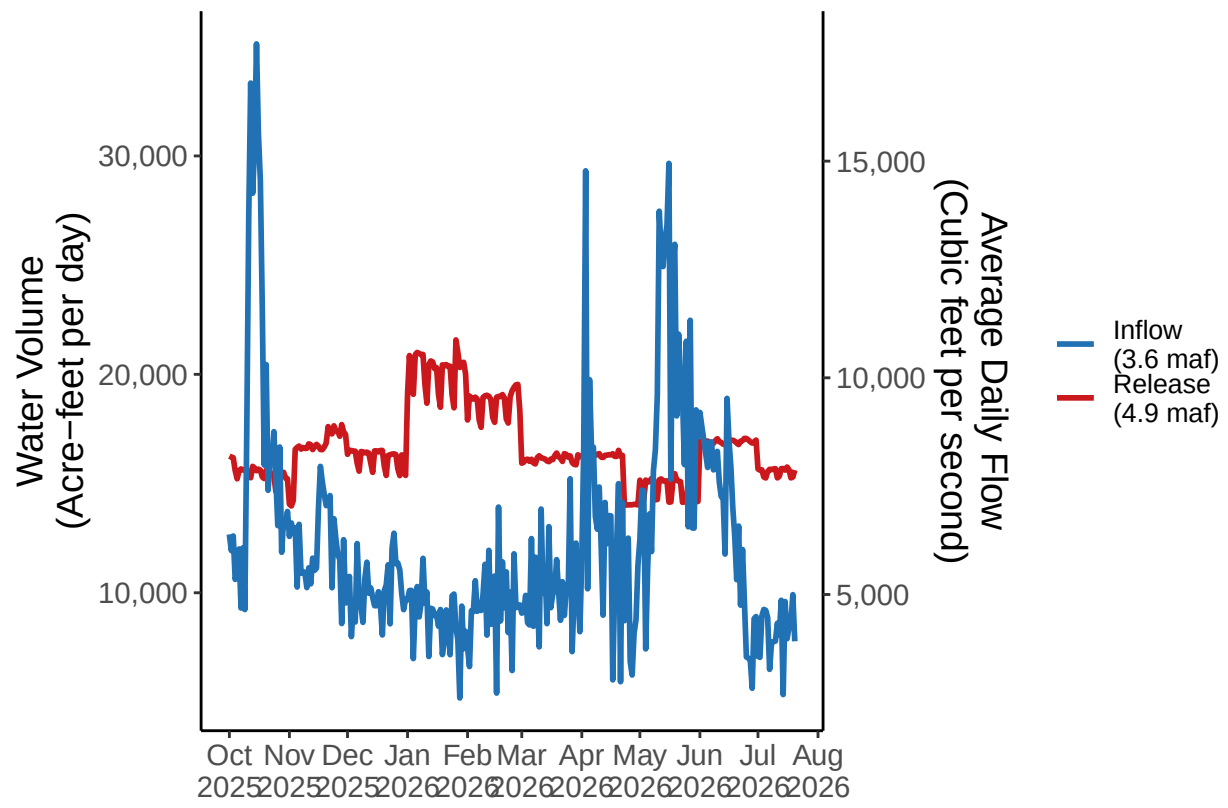


Figure 7: Frequency of Annual Lake Powell Release Volumes.

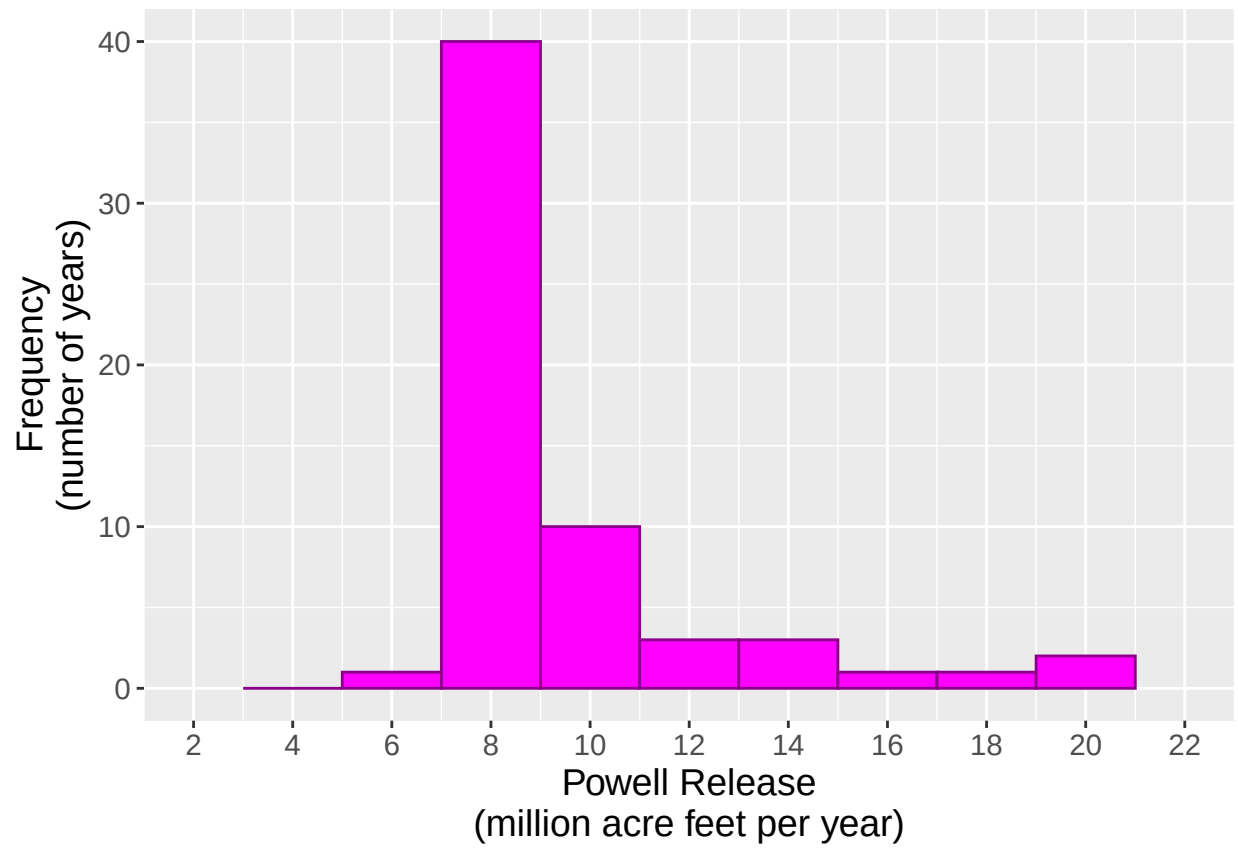


Figure 8: Frequency of Annual Lake Powell Inflow Volumes.

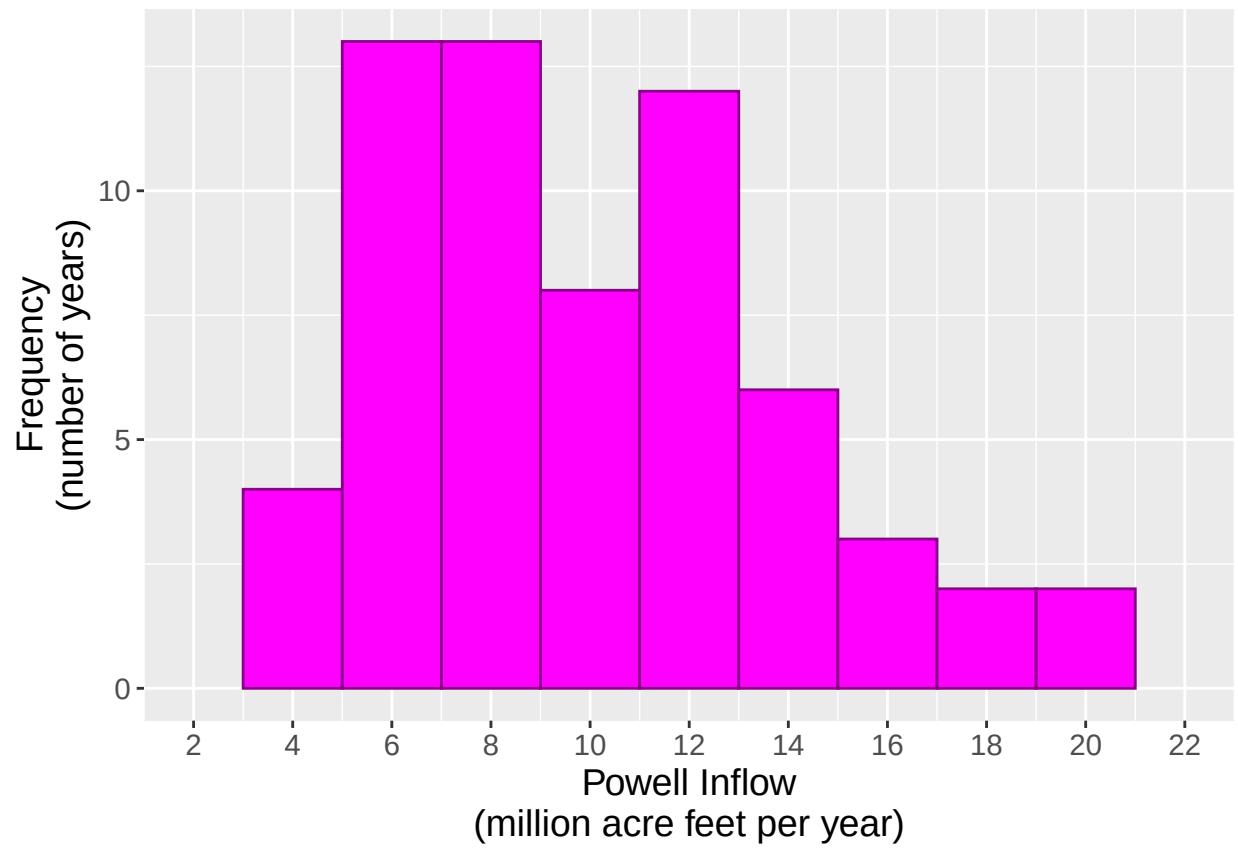


Figure 9: Lake Powell Inflows and Releases.

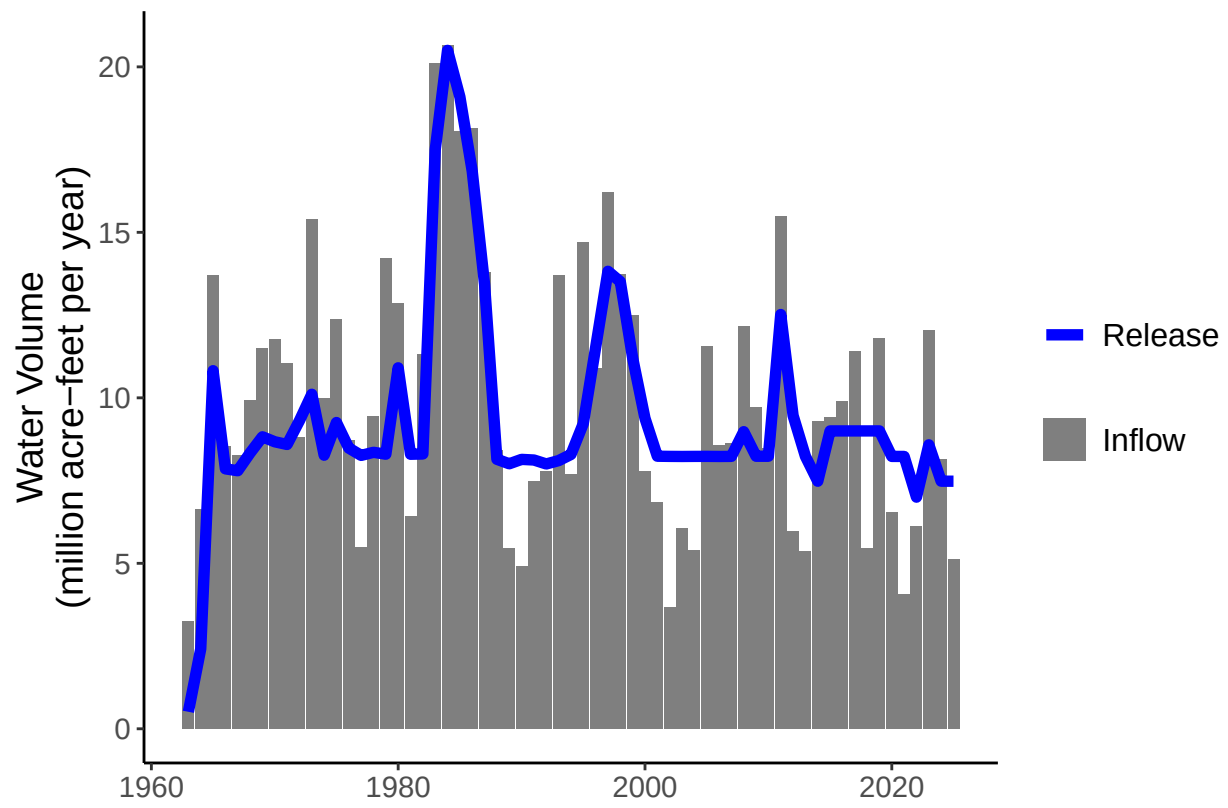


Figure 10: Annual Lake Powell Evaporation.

